

Introduction to AI along with Data Wrangling, EDA, Data Visulation

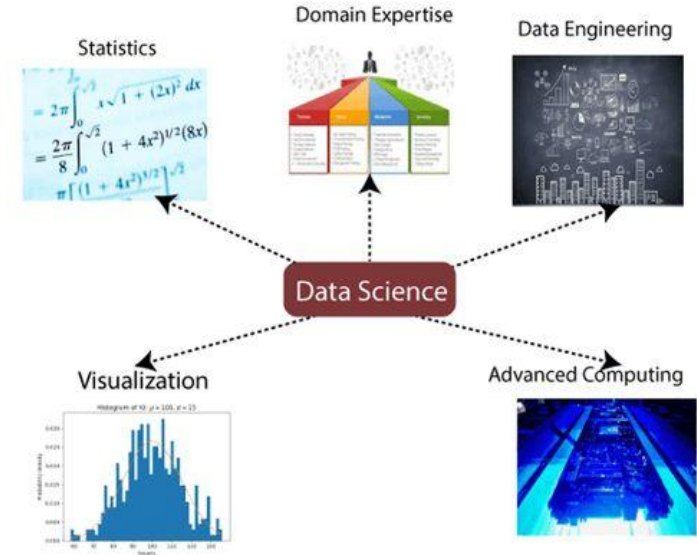
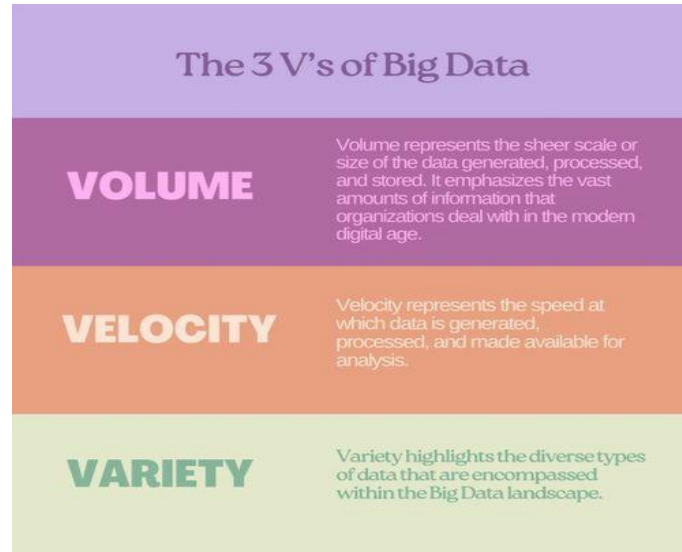
What is AI?

Artificial Intelligence (AI) is the simulation of human intelligence by machines, especially computer systems. It enables machines to perform tasks that typically require human intelligence—such as learning, reasoning, problem-solving, understanding language, and recognizing patterns.

What is Data Science?

Data Science is a field that combines statistics, computer science, and domain knowledge to extract meaningful insights from data.

What is Big Data?



Data Wrangling

Data wrangling, also known as data cleaning or data preprocessing, is the process of transforming and organizing raw data into a usable format for analysis.

Steps in Data Wrangling:

1. Discovery
2. Structuring
3. Cleaning
4. Enriching
5. Validation
6. Publishing

From the analysis of the Titanic Data set I got to learn about categorical data set and how to deal with it by either using Label Encoding and One hot Encoding.

Titanic Dataset

Findings and learning from the dataset:-

1. `df.isnull().sum()` is used to count the number of null rows in different columns in our data frame.
2. `df.duplicated().sum()` is used to count the number of duplicate entries in our df which is null in our dataset.
3. Handling Missing Values:

Missing value or null data are one of the most common problem while dealing with dataset. There are different approaches to deal with this problem and some of them are:-

- a. Imputation using Mean or Median or Mode
- b. Imputation using Group Statistics
- c. Dropping Missing Values

Handling with categorical columns:

- a. **Label Encoding** (Label encoding assigns a unique integer to each category.)
- b. **One-Hot Encoding** (One-hot encoding creates new columns indicating the presence (or absence) of each possible value in the original data.

Original

Island
Torgersen
Torgersen
Biscoe
Biscoe
Dream
Dream

```
df['Island'].unique()
```

```
['Torgersen', 'Biscoe', 'Dream']
```

Label Encoding

Island
0
0
1
1
2
2

```
le = LabelEncoder()  
le.fit_transform(df['Island'])
```

One-Hot Encoding

Torgersen	Biscoe	Dream
1	0	0
1	0	0
0	1	0
0	1	0
0	0	1
0	0	1

```
add_columns = pd.get_dummies(df['Island'])  
df = df.join(add_columns)
```

Matplotlib

Visualization library in Python.

Used for creating plots like Line, Scatter, Bar and Histogram.

Matplotlib is part of the standard Python data stack (pandas, NumPy, matplotlib, and Jupyter). pandas uses matplotlib as a backend to help visualize data in DataFrames.

Importing Matplotlib

```
import matplotlib
```

```
import matplotlib.pyplot as plt
```

Why pyplot?

Because pyplot is a submodule for creating interactive plots programmatically, and is often imported as the alias plt.

Two ways of creating plots

There are two main ways of creating plots in matplotlib:

1. `matplotlib.pyplot.plot()` - is recommended for simple plots (e.g. x and y)
2. `matplotlib.pyplot.XX` (where XX can be one of many methods, this is known as the object-oriented API) - is recommended for more complex plots (for example `plt.subplots()` can be used to create multiple plots on the same Figure, but we'll get to this later)

SYNTAX: `plt.plot();`

`plt.show()`

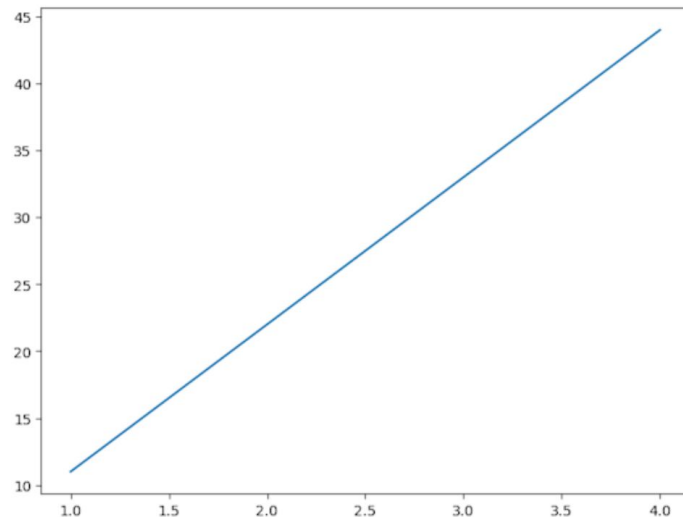
Creating a plot with the object-oriented version

`fig = plt.figure()`

`ax = fig.add_axes([1, 1, 1, 1])`

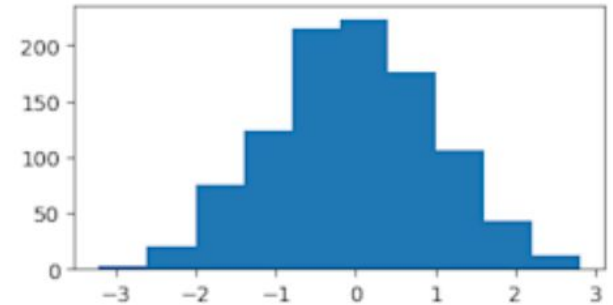
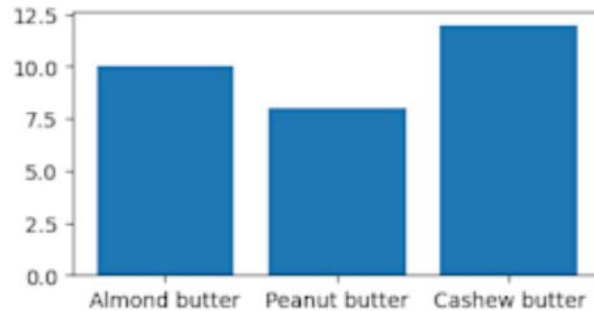
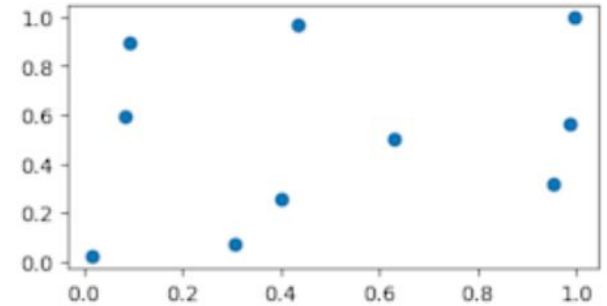
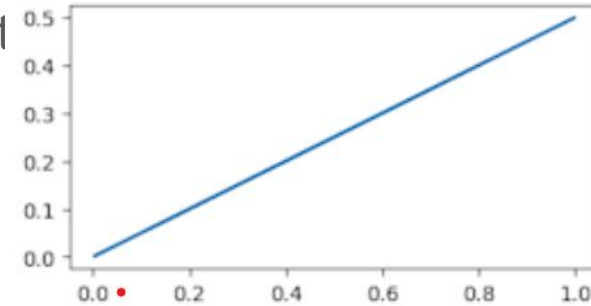
`ax.plot(x, y)`

`plt.show()`



Common types of plots using NumPy arrays:-

- Line plot - `ax.plot()` (this is the default plot in matplotlib)
- Scatter plot - `ax.scatter()`
- Bar plot - `ax.bar()`
- Histogram plot - `ax.hist()`



Exploratory Data Analysis (EDA)

EDA primary aim is to uncover underlying patterns, grasp the dataset's structure, and identify any potential anomalies or relationships between variables.

Terminologies

Distribution: The distribution of a dataset is how the dataset is spread out. We can visualize a dataset's distribution by observing its shape on a graph.

Outlier: An outlier is a data value that is significantly different, including much higher or lower, from the rest of a dataset.

Tools used in EDA

1. Pandas
2. Matplotlib
3. seaborn

Visualizing Distributions

Univariate Analysis

- Visualizing only one feature/variable at a time
- We use histplot for studying distribution of numeric values and countplot for categorical values.

Skewed Distribution

A skewed distribution is neither symmetric nor normal because the data values trail off more sharply on one side than on the other.

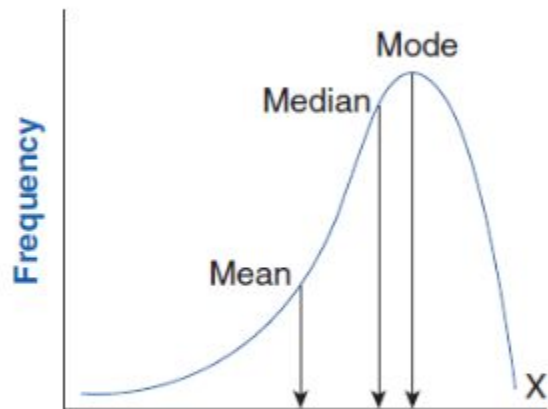
If one tail is longer than another, the distribution is skewed.

A left-skewed distribution has a long left tail.

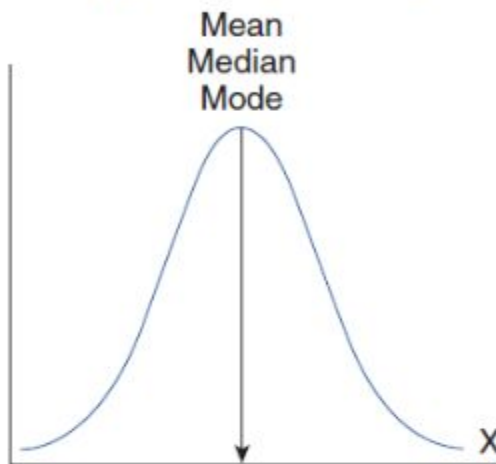
A right-skewed distribution has a long right tail.

Skewness can be observed using histogram or boxplot.

(a) Negatively skewed

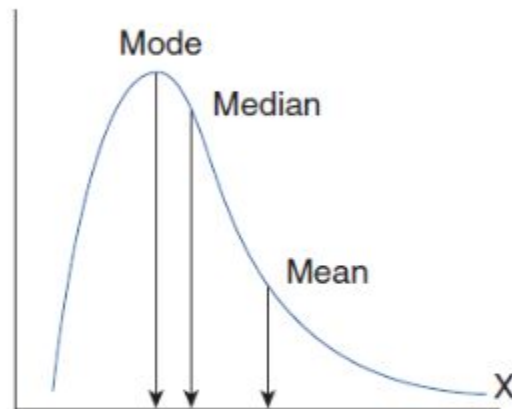


(b) Normal (no skew)



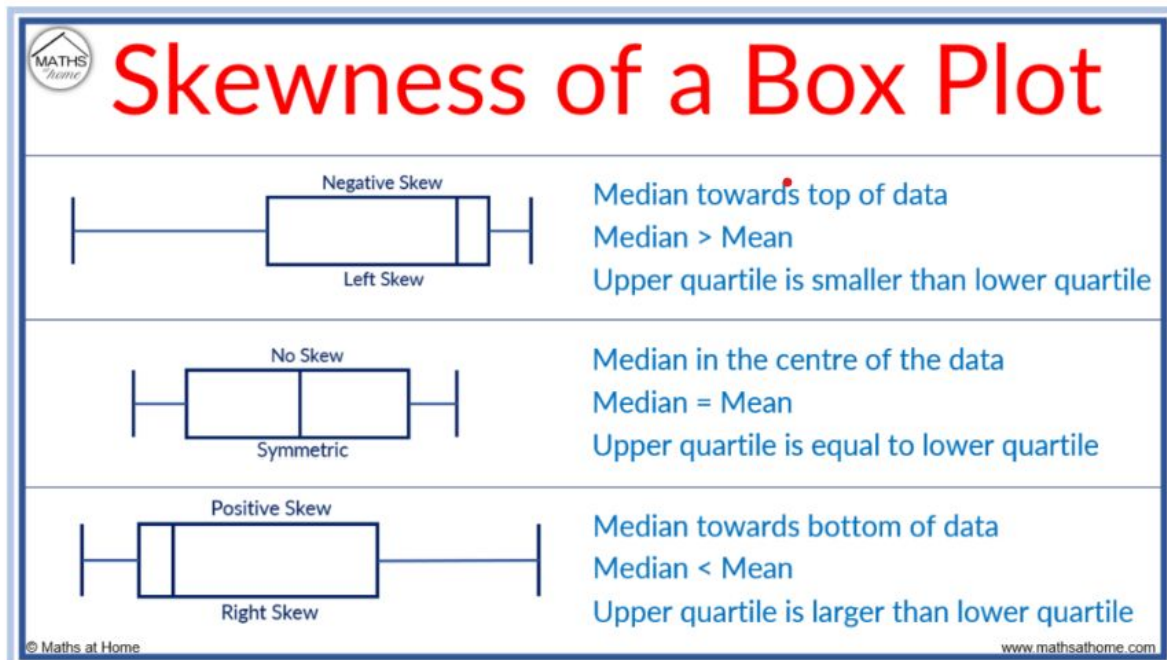
The normal curve
represents a perfectly
symmetrical distribution

(c) Positively skewed



Skewed Distribution

- A skewed distribution is neither symmetric nor normal because the data values trail off more sharply on one side than on the other.
- If one tail is longer than another, the distribution is skewed.
- A left-skewed distribution has a long left tail.
- A right-skewed distribution has a long right tail.
- Skewness can be observed using histogram or boxplot.



Correcting skewness using different transformation techniques:

- logarithmic transformation
- square root transformation
- cube root transformation
- Box-Cox transformation

Outliers can also skew the data. So, outliers removal can also solve the skewness.

The skewness value should be within the range of -1 to 1 for a normal distribution, any major changes from this value may indicate the presence of outliers.

Bivariate Analysis

Examine the relationships between different variables using scatter plots, box plots, and correlation matrices.

Box Plots

A box and whisker plot—also called a box plot—displays the five-number summary of a set of data. The five-number summary is the minimum, first quartile, median, third quartile, and maximum.

